



# Tran Pham Cong Danh

An AI Engineer: Embracing AI, Powered by Passion

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📍 HCMC, Vietnam

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## 1 WORK EXPERIENCE

- **AI Engineer** 12/2025 – Present  
**OSAS Joint Stock Company (via AI MED Technology Co., Ltd.)** HCMC, Vietnam  
**Description:** Primary AI Engineer driving the core AI capabilities of an enterprise chatbot product. Led the end-to-end design of ASR, TTS, RAG systems, and local LLM optimization. Key deliveries include: (1) a medical AI chatbot utilizing dynamic prompt classification for precise vector data routing; (2) a multi-level financial report chatbot with strict citation accuracy; and (3) multimodal R&D for skin disease detection and medical guidance.
  - **LLM & Infrastructure Optimization:** Optimized local LLM inference on a single RTX 5090 32GB using FlashAttention, KV cache quantization, and continuous batching. Validated via stress testing, the system successfully handled 15+ concurrent requests with extended context windows, achieving a 100% success rate (0% drop) with all responses returned within 12-20 seconds.
  - **Custom Memory Management:** Engineered a dynamic memory offloading engine to seamlessly transition between high-throughput/light-context and low-throughput/heavy-context workloads (up to 65k tokens), eliminating OOM errors and maximizing GPU utilization.
  - **ASR & TTS Pipelines:** Deployed CPU-optimized whisper.cpp for real-time transcription, augmented with a custom silence-detection module to suppress hallucinations; integrated Piper TTS with a 1,000+ term medical pronunciation dictionary.
  - **Medical AI & Multimodal:** Trained RF-DETR XXL for brain tumor segmentation, achieving a validation DICE score of 91.77% and an mIoU of 84.32%. Developed a SigLIP2-based multimodal retrieval system for skin disease classification using image-text embeddings.

*Note: Work projects are proprietary; code and detailed metrics available upon request during interview.*
- **AI-Driven Marketing Supporter** 04/2025 – 05/2025  
**Karuna Veggie Restaurant** HCMC, Vietnam
  - Drove AI-powered marketing to increase brand visibility and customer engagement.
- **Web Developer** 01/2021 – 12/2021  
**Dr. Nhung Dental Clinic** HCMC, Vietnam
  - Architected a PHP-MySQL internal management portal to streamline patient records and revenue tracking.
  - Built a responsive UI with secure database optimization and a jQuery-powered image gallery, improving usability and operational efficiency.

## 2 TECHNICAL SKILLS

- **Programming Languages:** Python, JavaScript, PHP, C++
- **AI, LLM, ASR & TTS:** PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face, LangChain, Ollama, RAG Architecture, Qdrant, Vector Search, Prompt Engineering, whisper.cpp (ASR), Piper TTS (TTS)
- **Computer Vision & Model Optimization:** OpenCV, RF-DETR, SigLIP2, ONNX, FlashAttention, KV Cache
- **Data Management & Processing:** SQL, NoSQL, Apache Spark, Kafka, Jupyter
- **System & DevOps Tools:** Git, Linux, Docker, MLOps
- **Python Libraries & UI:** NumPy, Pandas, Matplotlib, Gradio, Streamlit, Flask
- **Web & Other Tools:** HTML, CSS, Selenium, System Design

### 3 PERSONAL PROJECTS

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- **BTCUSDT ML ETL Pipeline** | Python, Docker, PostgreSQL, MinIO, Spark, DaskDB 2025
  - Scalable ML ETL pipeline for BTC-USDT data processing and LSTM price forecasting.
- **ViHealth Chat LoRA** | PyTorch, Hugging Face, LoRA 2025
  - Fine-tuned LLM with LoRA on Vietnamese medical textbooks for accurate, contextual healthcare responses and rehabilitation guidance.
- **Vi-Whisper Tiny Finetuning** | Whisper, Gradio, Python 2025
  - Lightweight, real-time Vietnamese ASR model optimized for local deployment with Gradio/Python interfaces.
- **SlimFace & Anime Super Resolution** | OpenCV, Real-ESRGAN, Transfer Learning 2025
  - Developed efficient face recognition for edge devices and enhanced anime-style image clarity using fine-tuned Real-ESRGAN.

### 4 EDUCATION

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- **Master of Engineering in Computer Science** 2022 – 2024  
Ho Chi Minh City University of Technology – Vietnam National University Ho Chi Minh City *GPA: 8.08/10*
  - **Thesis:** Classification of blood pressure for patients treated at home.
- **Doctor of Medicine (General Medicine)** 2014 – 2020  
University of Health Sciences – Vietnam National University Ho Chi Minh City *HCMC, Vietnam*

### 5 CERTIFICATES

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- **Deep Learning in Python** – DataCamp (Nov 2024)
- **Python Developer** – DataCamp (Sep 2025)
- **Machine Learning Engineer** – DataCamp (Nov 2024)
- **Python Essentials 2** – NetAcad (Sep 2024)
- **OpenCV Bootcamp** – OpenCV (Sep 2024)
- **Machine Learning with Python** – freeCodeCamp (Sep 2024)